



資料結構

Data Structure

Lab 10

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Lab10-Ex1.

From the three given input.txt files, randomly select one file. Based on the elements in the selected file, hand-draw the step-by-step construction process of both a Max Heap and a Min Heap. Diagrams must be provided for every step.

Result

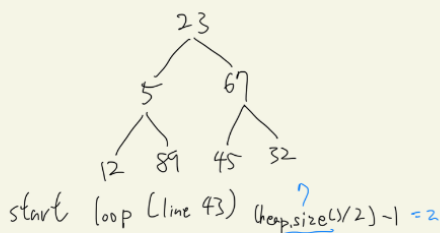
File: input2.txt

Application used: Prodrafts on iPad

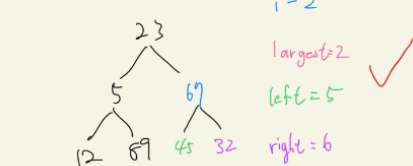
Max Heap construction process:

input 2 arr = 23, 5, 67, 12, 89, 45, 32

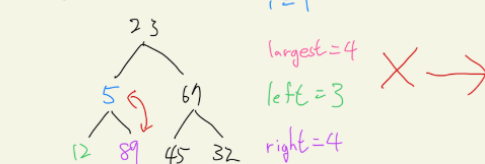
buildMaxHeap(arr)



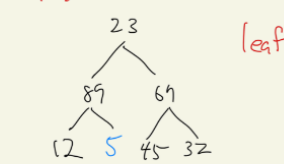
heapify(2)



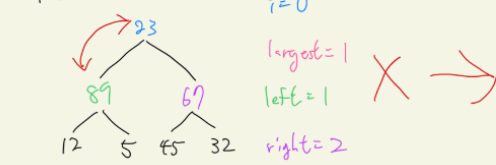
heapify(1)



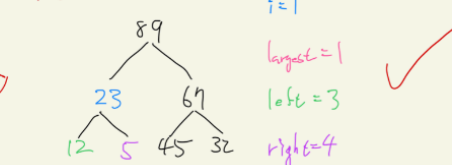
heapify(4)



heapify(0)

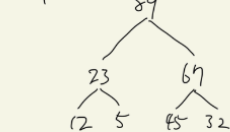


heapify(1)



loop end

heap

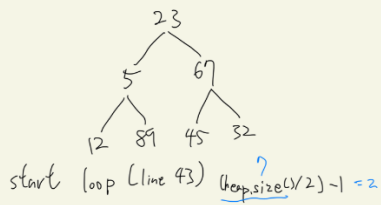


output: 89 23 67 12 5 45 32

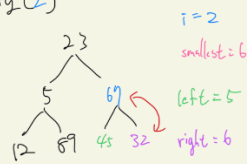
Min Heap construction process:

input: arr = [23, 5, 67, 12, 89, 45, 32]

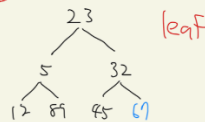
build MinHeap(arr)



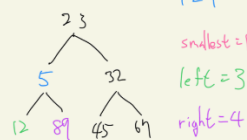
heapify(2)



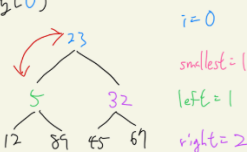
heapify(6)



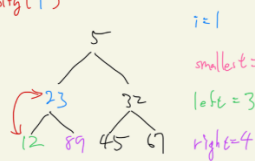
heapify(1)



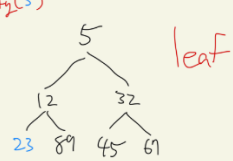
heapify(0)



heapify(1)

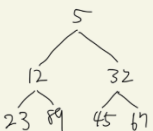


heapify(3)



loop end

heap



output: 5 12 32 23 89 45 67

Lab10-Ex2.

Please implement the missing heapify function for the Max Heap in the sample code, ensuring that both Max Heap and Min Heap work correctly. Then, read the three input files (input1.txt, input2.txt, and input3.txt) to test your output results.

Code

```
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
using namespace std;

// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
    vector<int> arr;
    ifstream file(filename);

    if (!file) {
        cerr << "Error opening file: " << filename << endl;
        return arr;
    }

    string line;
    while (getline(file, line)) { // 持續讀取整行內容
        stringstream ss(line);    // 創建字符串流
        string value;
        while (getline(ss, value, ',')) { // 用逗號分隔值
            try {
                arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
            }
            catch (exception& e) {
                cerr << "Invalid number format in file: " << value << endl;
            }
        }
    }

    file.close();
    return arr;
}

class MaxHeap {
public:
    vector<int> heap; // 儲存Max Heap的元素

    // 建立Max Heap
    void buildMaxHeap(vector<int>& arr) { // 建立Max Heap
        heap = arr;

        for (int i = (heap.size() / 2) - 1; i >= 0; i--) { // 從最後一個非葉子節點
            開始向上執行Max Heap
            heapify(i);
        }
    }
}
```

```

void heapify(int i) {// 堆化函式 (確保以 i 為根的子樹符合Max Heap性質)
    int largest = i; // 假設當前節點是最大的
    int left = 2 * i + 1; // 左子節點索引
    int right = 2 * i + 2; // 右子節點索引

    // if判斷式, 檢查左子節點是否為有效範圍且比當前節點 (父節點) 大
    if (left < heap.size() && heap[left] > heap[largest]) {
        largest = left; // update largest to left if left child is within
range and holds a greater value
    }

    // if判斷式, 檢查右子節點是否為有效範圍且比當前節點 (父節點) 大
    if (right < heap.size() && heap[right] > heap[largest]) {
        largest = right; // update largest to right if right child is
within range and holds a greater value
    }

    // if判斷式, 如果最大的不是父節點, 交換並繼續堆化
    // 遞迴處理受影響的子樹
    if (largest != i) {
        swap(heap[i], heap[largest]); // swap the current node with the
largest child node found
        heapify(largest); // recursively heapify to restore
heap properties
    }
}

// 顯示Heap的內容 (使用BFS)
void printHeap() {
    for (int val : heap) {// 遍歷 Max Heap中的每個元素
        cout << val << " ";// 輸出元素
    }
    cout << endl;
}

};

int main() {
    // 從文件讀取輸入元素
    string filename = "input3.txt"; // 請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信息
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
    maxHeap.buildMaxHeap(arr); // 建立Max Heap
}

```

```
// 輸出Max Heap的內容
cout << "Max Heap: ";
maxHeap.printHeap();
cout << endl;

system("pause");
return 0;
}
```

Result

```
Microsoft Visual Studio Debug Console
Input Array: 10 5 15 3 7 12 18
Max Heap: 18 7 15 3 5 12 10

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

輸入Input1.txt後Max heap輸出結果

```
Microsoft Visual Studio Debug Console
Input Array: 10 5 15 3 7 12 18
Max Heap: 18 7 15 3 5 12 10

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debug Console
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 12 10 7 15 18

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

輸入Input1.txt後Min heap輸出結果

```
Microsoft Visual Studio Debug Console
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 12 10 7 15 18

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debug Console
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 23 67 12 5 45 32

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

輸入Input2.txt後Max heap輸出結果

```
Microsoft Visual Studio Debug Console
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 23 67 12 5 45 32

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debug Console
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 32 23 89 45 67

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

輸入Input2.txt後Min heap輸出結果

```
Microsoft Visual Studio Debug Console
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 32 23 89 45 67

Press any key to continue . . .

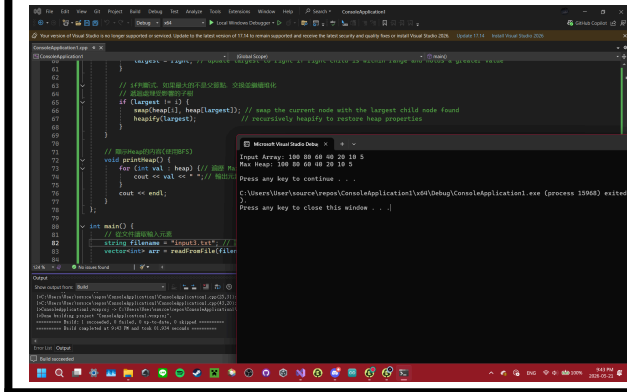
C:\Users\User\source\repos\ConsoleApplication1\bin\Debug\ConsoleApplication1.exe
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debug Console
Input Array: 100 80 60 40 20 10 5
Max Heap: 100 80 60 40 20 10 5

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\...
Press any key to close this window . . .
```

輸入Input3.txt後Max heap輸出結果



```
Microsoft Visual Studio Debug Console
Input Array: 100 80 60 40 20 10 5
Min Heap: 5 20 10 40 80 100 60

Press any key to continue . . .

C:\Users\User\source\repos\ConsoleApplication1\...
Press any key to close this window . . .
```

輸入Input3.txt後Min heap輸出結果

